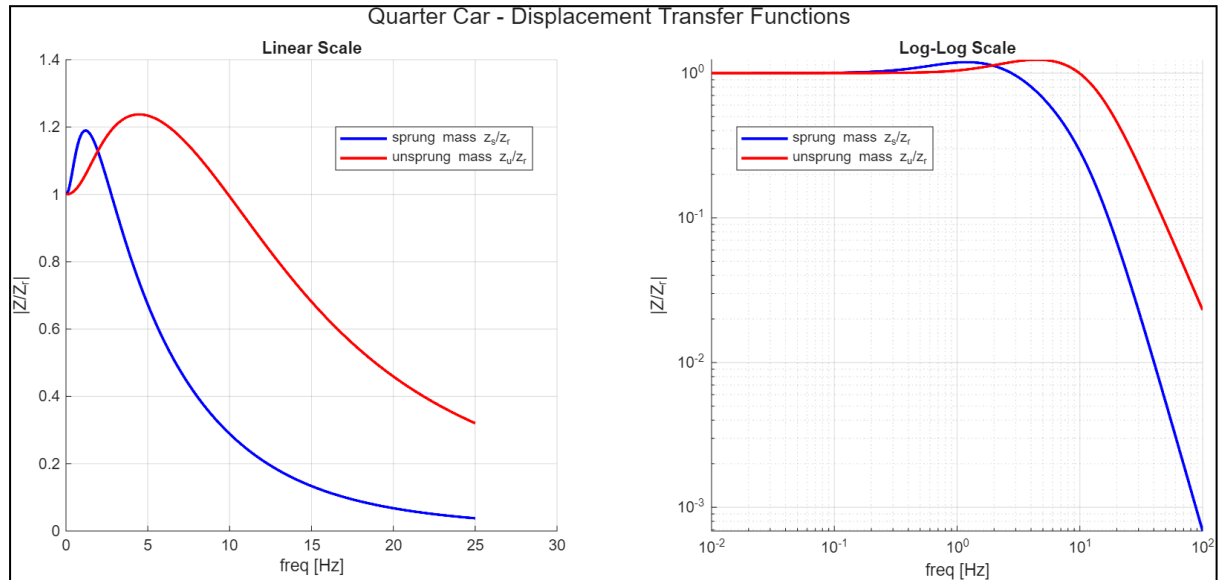


VEHICLE DYNAMICS AND CONTROL- Vertical Model

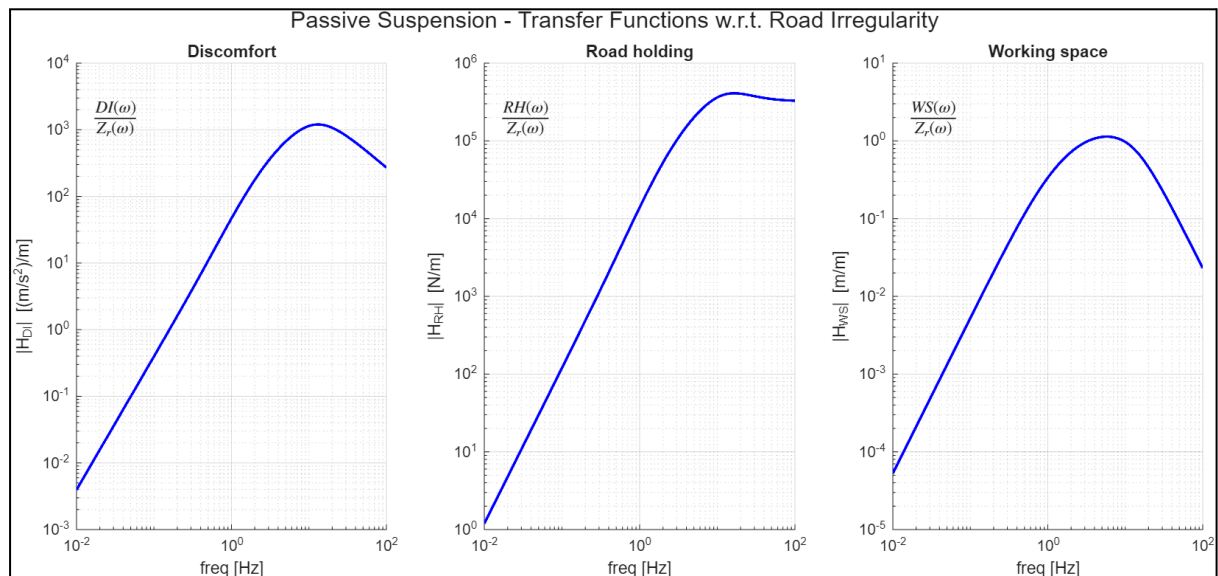
CAR SEGMENT - Sports Car

A. Finding the best Damping Coefficient for Front and Rear

- Displacement TFs for Sprung and Unsprung Mass

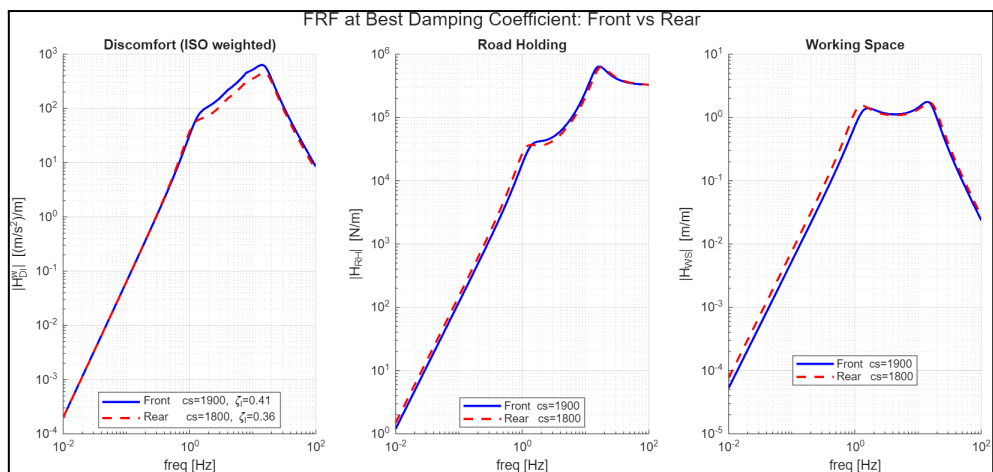
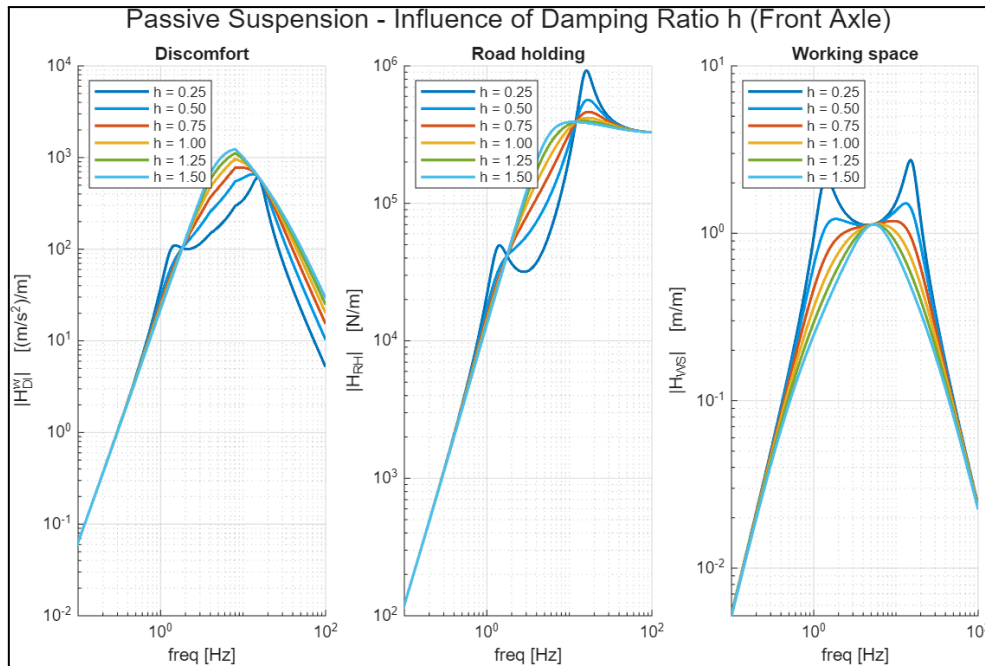


- TFs wrt Road Irregularity

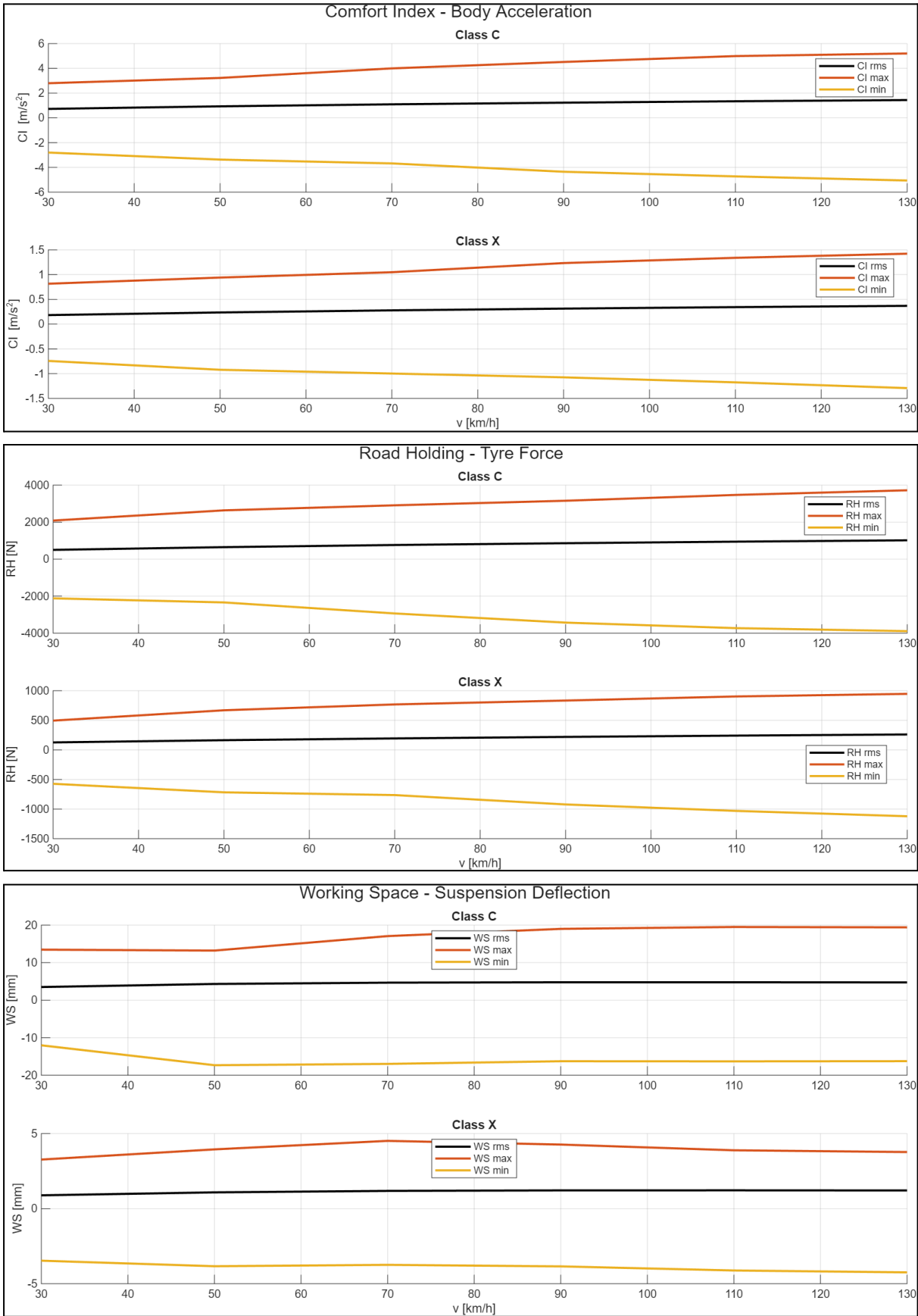


- Ideal Damping Values

=== CRITICAL DAMPING VALUES ===				
Axle	cc_I [Ns/m]	cc_II [Ns/m]	wn_I [Hz]	wn_II [Hz]
Front	4628.4	6899.3	1.375	15.686
Rear	5000.0	6368.7	1.146	16.893
=== BEST DAMPING COEFFICIENTS ===				
Axle	cs [Ns/m]	zeta_I	zeta_II	
Front	1900	0.411	0.275	
Rear	1800	0.360	0.283	

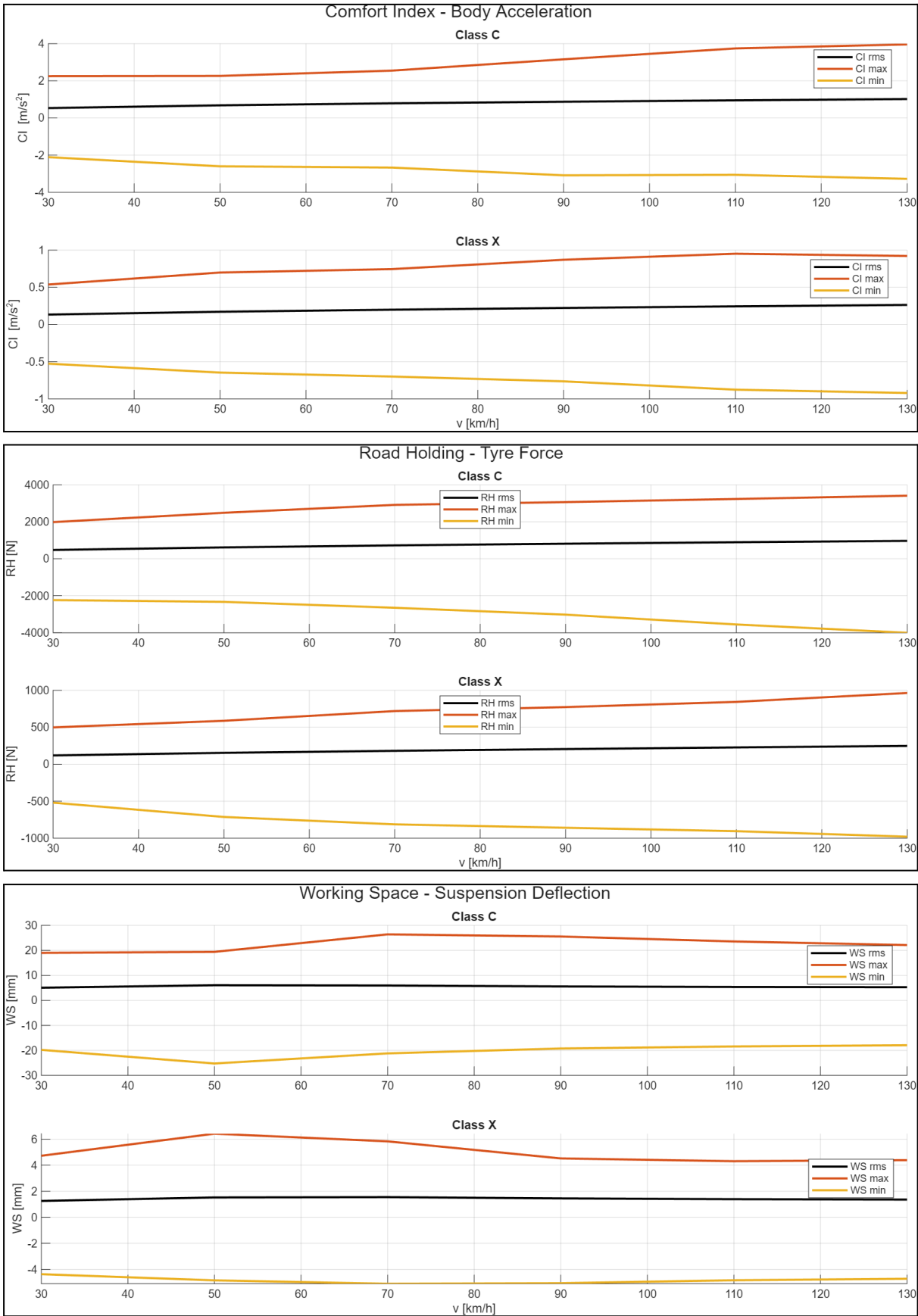


B. Dynamic Simulation on Passive Damping

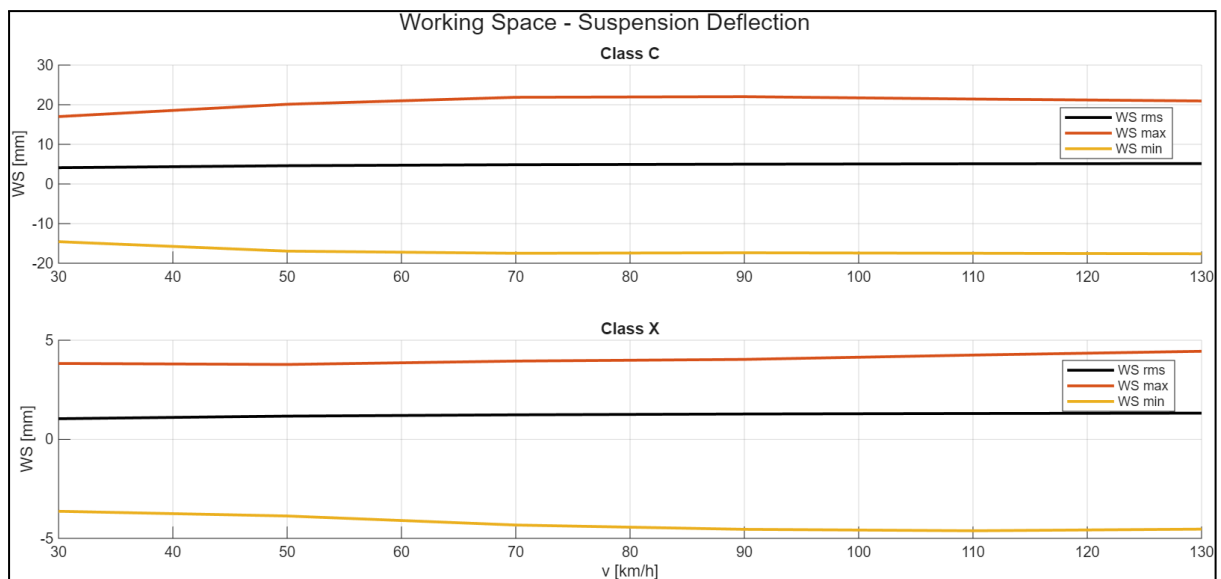
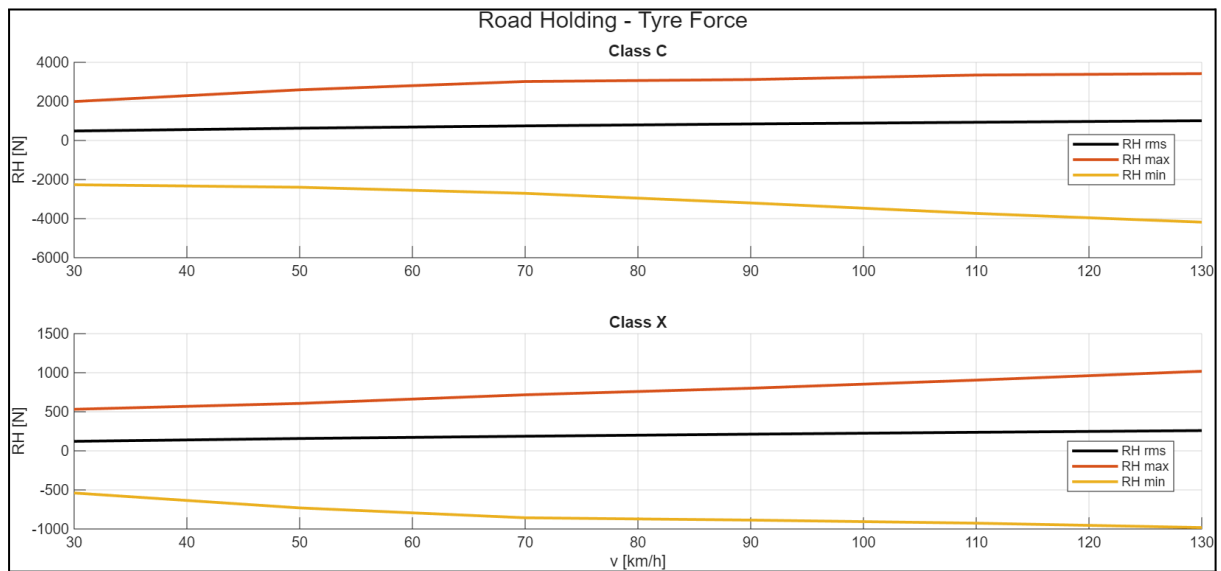
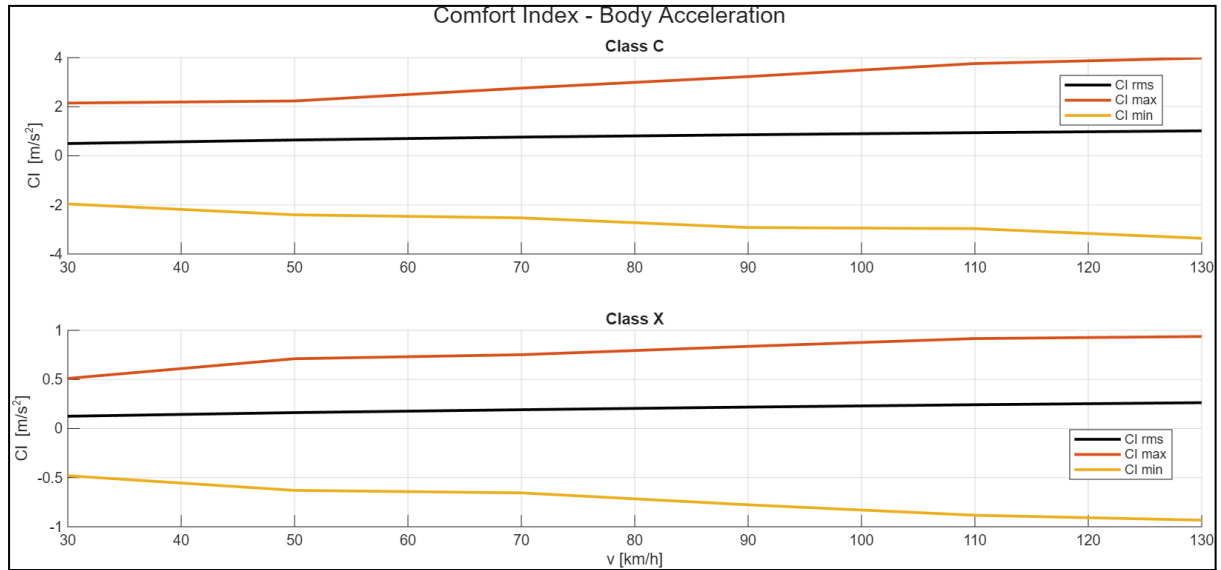


C. Dynamic Simulation on Semi-Active and Active Damping

● Semi-active Damping



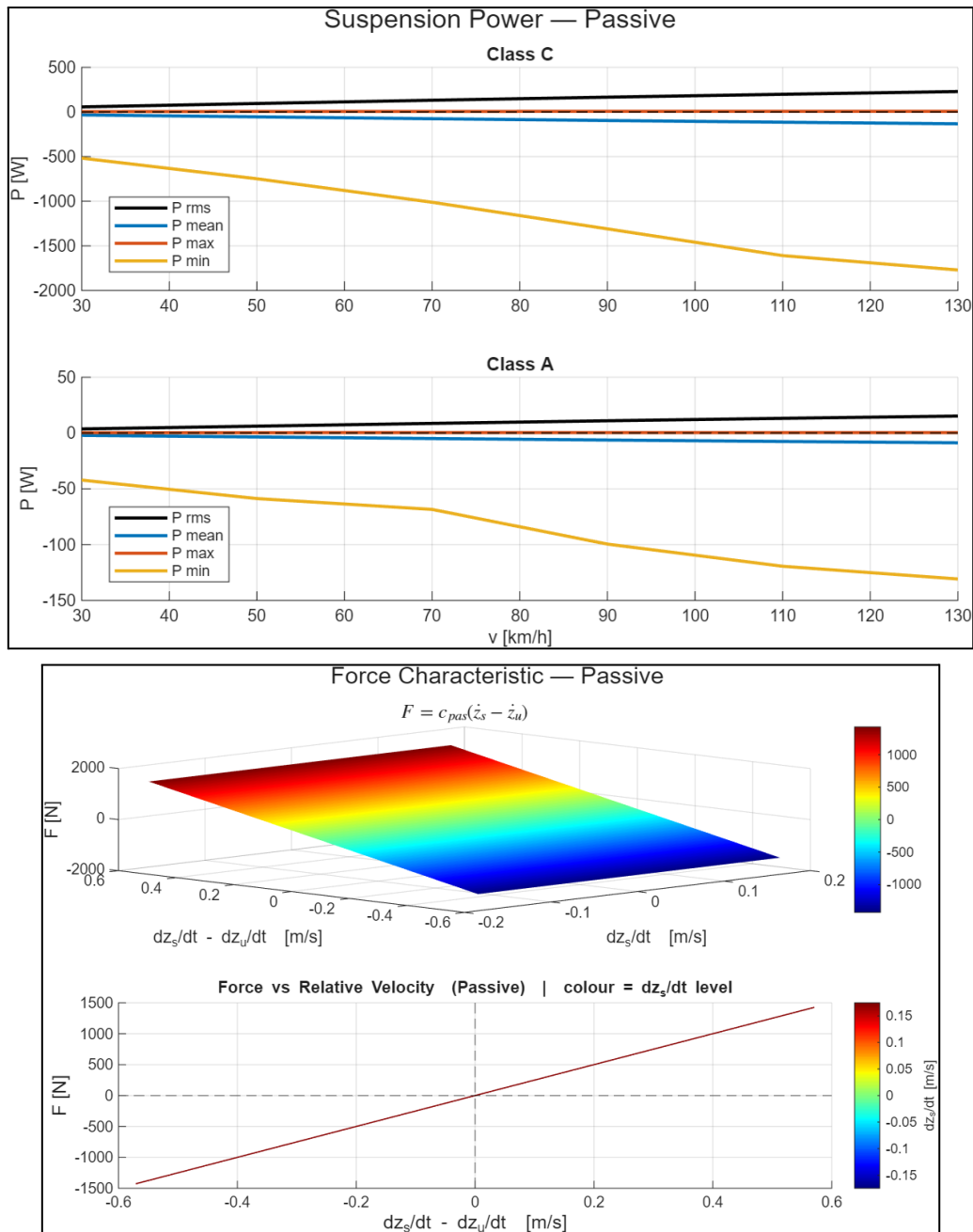
- **Active Damping**



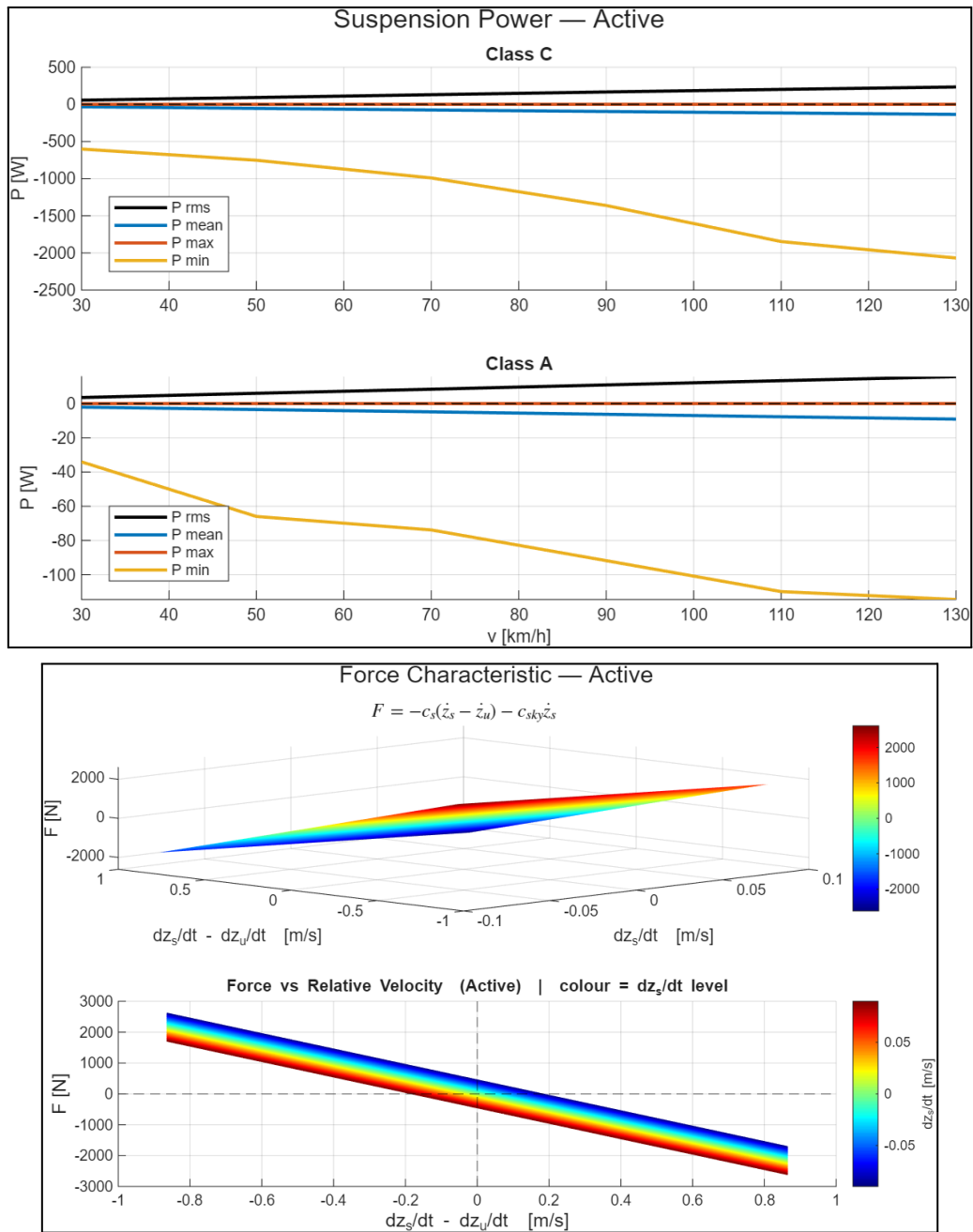
The dynamic simulations on the passive, semi-active and active suspensions highlight the various advantages and limitations of these dampers. The passive suspension exhibits the largest tyre force variations and body acceleration levels across the investigated speed range. Introducing controlled damping significantly improves vehicle performance, reducing dynamic tyre load fluctuations and slightly improving ride comfort. The semi-active suspension achieves most of these improvements, while the active suspension provides only marginal additional benefits. Consequently, the largest performance gain is observed when moving from passive to semi-active control, whereas the difference between semi-active and active suspensions remains relatively small for the analysed operating conditions.

D. POWER DISSIPATION BY PASSIVE SUSPENSION AND ACTUATOR

i. Passive Damping



ii. Active Damping



iii. Semi-Active Damping

